

CS676-Project1 Report

This report describes the credibility scoring system project. The baseline scoring system evaluates a URL's trustworthiness using two layers: a rule-based layer that inspects the URL string and an LLM-based layer that uses Claude calls to provide context and support judgment. At the baseline, the two layers are blended using a fixed split weight that contributes to a final score: `RULE_WEIGHT = 0.6` and `LLM_WEIGHT = 0.4`, and the system intentionally runs on weaknesses to be addressed to improve performance, including inability to distinguish a preprint from a peer-reviewed paper, no handling of retracted papers, any .edu scores highly even if it's a student's personal homepage, and an unjustified constant blend weight between the two layers. This report documents improvements made, measures the effect using the provided [evaluate.py](#) scoring, and discusses what still fails in the credibility scoring. Note that the report and project are based on `JUDGE_MODEL` set to Claude Opus 5 due to it being the most capable model for best performance.

The rule layer is a hand-written guess about what makes a source credible; thus, the same URL always produces the same score. The limitation is that it can only recognize domains explicitly entered in the lookup table, and any unknown domain falls back to coarse `TLD_SCORES`. The LLM layer's advantage is the reverse: because the underlying model was trained on a broad range of real-world knowledge, it can reason about domains never explicitly listed. For example, distinguishing that `jamanetwork.com` belongs to a peer-reviewed medical journal even though no rule was ever written for it. Its costs are latency, API expense, and reduced determinism compared to the rule layer.

Defect: Preprints Scored Identically to Peer-Reviewed

The baseline assigned [arxiv.org](#) and [biorxiv.org](#) domain scores of 0.75 and 0.70 – making both preprint domains too close to established peer-reviewed journals (0.92-0.95). I believe these represent two distinctive types of sources: a publication that is reviewed rigorously, and an archived document that can range from being highly cited to being completely unreviewed.

Accordingly, I further reduced both domains' values to account for this distinction in domains: [arxiv.org](#) from 0.75 to 0.62, and [biorxiv.org](#) from 0.70 to 0.55. These values were tuned empirically against the evaluation set's labeled examples using [evaluate.py](#), which also encodes the instructor's judgment that arXiv (expected 0.65) and bioRxiv (expected 0.50) are to be trusted differently as well. Likely, similar to my approach, arXiv

could be slightly more trustworthy than bioRxiv due to its endorsement system making a higher barrier to entry to publish, and also its longer history.

This change reduced both arXiv's individual scoring error from 0.12 to 0.01 and bioRxiv's from 0.32 to 0.17, and contributed to an overall Mean Absolute Error reduction from 0.142 to 0.131, with band accuracy rising from 66.7% to 75%.

Defect: .edu Domains Scored Uniformly Regardless of Content

The baseline's TLD fallback assigns any unrecognized .edu domain a score of 0.82, treating it as trustworthy. However, it is important to note the possibility of the limitation of failing to distinguish an official university publication from a student homepage or publication hosted on the same domain.

To address this, I added a new signal, `edu_student_homepage`, that detects a `/~` path pattern on .edu domains indicating a possible student directory and assigns it a penalty of -0.42, positioning it similarly to other self-published content in `DOMAIN_SCORES`. I believe this clarifying addition is reasonable, as a personal homepage on a university server likely does not carry the same rigorous editorial oversight as institutional-grade credibility (0.82); thus, I judged it to be similar to other self-published credibility scores.

A corresponding test case was added to `test_credibility.py`, testing the new signal against a synthetic URL, <https://www.university.edu/~studentxyz/mypublication>, confirming that the clarifying fix produces a penalty from the baseline 0.82 credibility score for any unrecognized .edu domain. The test case verified that an official .edu publication outranks a .edu student publication, and that the student publication scores comparably to self-published content. However, since the current 24-URL evaluation set in [evaluate.py](#) does not include a .edu student-page example, there was no effect on the reported MAE.

Defect: No Handling of Retraction or Real Citation Awareness

The baseline scoring system had no mechanism to distinguish a highly-cited, credible publication from a retracted or unverified new publication. For example, the arXiv and bioRxiv URLs were scored purely by domain, regardless of the specific publication's real-world context. To address this, a new signal, Signal 6, was added that extracts a DOI from the URL path and uses the OpenAlex API to retrieve the specific publication's field-weighted citation impact (FWCI) and retraction status. I chose FWCI over a raw citation count because citation counts vary across fields – a specialized research area naturally accumulates far fewer citations compared to a fast-moving, larger-scale field.

FWIC handles this and normalizes citations against the average for its specific field, where a value of 1.0 represents the field's average. This provides more accurate context for credibility scoring than citation count alone.

The created signal applies a -0.50 penalty for retracted publications, a -0.10 penalty for below-average field-normalized impact (FWCI < 0.5), a neutral adjustment of 0.0 for typical average impact (FWIC between 0.5 and 1.5), and a bonus of +0.08 for high field cited-work (FWCI >= 1.50).

If there is no DOI, the API is inaccessible, or the DOI is not indexed in OpenAlex, the call fails gracefully, and scoring falls back to the baseline domain/TLD-based system.

This was independently verified against a real cited publication, 10.1056/NEJMoa2034577, a New England Journal of Medicine article, correctly producing a high_impact signal.

However, it returned a credibility score of 1.0 and was capped by _combine_signals, indicating the baseline's further weakness in calibration (defect #7) and the additive, rather than statistical, logic of _combine_signals (defect #6).

This test does not affect the MAE provided on the 24-URL evaluation set because the set contains synthetic placeholders that are not in the OpenAlex indexed works.

Furthermore, a limitation to the creation of signal 6 is that it only works when the DOI is present in the URL itself, which some URLs don't include, such as JAMA's article. To attempt to address this, a Signal 7 was introduced to look at the actual webpage and fetch the citation_doi meta tag. The signal worked as intended; however, JAMA's site blocked the attempt to look at its page because of bot detection in this particular case, so the JAMA credibility score on the evaluation set with the LLM off produced no improvements in the worst single error.

Defect: Unjustified Constant Blend Weight

The blend weight between the two layers is one of the most influential parameters in the credibility system, and the baseline's choice of RULE_WEIGHT = 0.6 was never tested to determine if it was a good split.

The rule-layer improvements described above led to a reduced MAE from 0.142 to 0.131 and improved band accuracy from 66.7% to 75%, but the worst single error stayed at 0.410. The signal to attempt to address the worst single error (signal 7) could not improve it in this case due to bot detection in the attempt to fetch the meta tags.

I tested a split of RULE_WEIGHT = 0.3 and LLM_WEIGHT = 0.7 against the evaluation set. The summary of the evaluation produced an MAE = 0.058, band accuracy = 91.7%, and a worst single error of 0.130, a big improvement over the baseline split (RULE_WEIGHT = 0.6, LLM_WEIGHT = 0.4), which produced MAE = 0.142, band accuracy = 66.7%, and a worst single error of 0.410.

An intermediate split, RULE_WEIGHT = 0.5, LLM_WEIGHT = 0.5, was also tested. Producing an MAE = 0.072 and band accuracy = 91.7% – marginally worse than the 0.3/0.7 split.

	RULE_WEIGHT	LLM_WEIGHT	MAE	Band Accuracy	Worst Error
BASELINE	0.6	0.4	0.142	66.7%	0.410
Rule Improvements (LLM off)	0.6	0.4	0.131	75%	0.410
Rule Improvements (LLM on)	0.6	0.4	0.082	87.5%	0.230
Weight Blend test 1	0.5	0.5	0.072	91.7%	0.190
Weight Blend test 2	0.3	0.7	0.058	91.7%	0.130

RULE_WEIGHT = 0.3 was chosen as the set split for two reasons. First, it produced the best credibility performance and no indication of possible overfitting. Second, this choice makes sense given that the weaknesses of the credibility system were coming from the short, hand-written held-out block. Giving the LLM layer more weight allows the model to use real-world context to carry more of the judgment where the domain tables fail.

More test splits across additional values were not completed due to my system's disk space constraints encountered throughout the project; however, the three split tests above nonetheless do show measured improvement and that the baseline default of 0.6 and 0.4 were not optimal, and that letting the LLM get more weight indicates the positive direction.

Defect: Explanation Quality

The baseline's explanation field was built by just gluing together every signal's raw reason with semicolons, and although it provides the user with what rules were triggered, it fails to give the user a sense of why the result should be trusted.

A new function, `credibility_explanation()`, was made to replace the weak explanation with a purpose-written sentence that would explain the source's significance. The function starts with the basic domain judgement, then adds a reasonable explanation of why the source is credible. For example, the NEJM article now produces: *"'nejm.org' is a domain we recognize. This specific work is cited far more than average for its field (297.97× the typical rate), indicating strong influence and scrutiny within the research community."*

The function involved ranking the signals by how useful they're to the user. My first iteration involved picking the "largest" value as the most important signal – but that resulted in the function picking a reason not as purposeful for the user, such as the *"URL has a DOI..."*, instead of a more meaningful one like citation impact, simply because its number happened to be bigger. To address this, I ranked signal types by how informative they were to the potential user; for instance, retraction status and citation impact are prioritized over a minor technical detail like the presence of a DOI.

LITERATURE REVIEW

Trade-offs between automation, cost, and rigour exist across different approaches to source credibility assessment. Credibility assessment systems draw on a combination of academic signals, including peer review or institutional publication, citation signals, and metadata signals like author credentials and publication dates. My approach to credibility assessment relies on the combination of a rule-based layer and OpenAlex integration to capture academic and citation-based signals.

NewsGuard is a commercial credibility rating service consisting of a team of journalists and experienced editors that evaluate news and information sites against nine journalistic criteria to produce a site score out of 100 (<https://www.newsguardtech.com/ratings/rating-process-criteria/>). The approach to credibility scoring is apolitical and thorough, with a clear process; however, I believe it could be limited in terms of scalability since it relies on manual evaluation, leading to missing new or niche sources.

Wikipedia takes a community-based approach to source credibility. The trustworthiness of a source is determined through open discussions. Furthermore, Wikipedia also classifies preprints as generally lower in credibility, labeling them as “seldom reliable sources”, compared to peer-reviewed sources, validating my approach to lowering their score in the credibility system. Wikipedia’s credibility assessment takes into account citation count as a credible signal – checking a source’s citation count as a way to confirm if it has entered mainstream academic discourse (https://en.wikipedia.org/wiki/Wikipedia:Reliable_sources). This produces a well-defended judgement for credible scoring for sources the community has actually discussed, but its limitation falls on sources that have never come up.

OpenAlex has a distinctive approach to credibility, and they state their philosophy deliberately is to impose “no quality bar” on which sources it indexes. They believe that there’s utility in indexing everything, including “lower-quality” sources, as they help fuel high-power meta-analysis (<https://help.openalex.org/data/sources/#no-quality-bar-by-design>). OpenAlex documentation also notes that “lower-quality” work can always be filtered out if it’s included – it can’t be added back if it’s not. Rather than using a credibility score, OpenAlex relies on “allow lists,” curated lists of trusted sources, and prefers them over “deny lists” because they’re easier to maintain and more transparent. On the other hand, this differs from the project’s credibility system, which works on a single handwritten allow list (DOMAIN_SCORES) with fixed scores.

This credibility system’s hybrid rule-and-LLM approach draws from OpenAlex’s citation-based reasoning, while relying on a simpler, single allow list rather than NewsGuard’s manual review or Wikipedia’s community open discussions.

Limitations

The project fixed several of the defects listed, including preprint scoring, .edu student pages, retraction/citation awareness through OpenAlex, the understanding of the blend split, and improved explanation quality for the user. An attempt to improve defect #1 was also made; although it worked on [Nature.com](https://www.nature.com), it did not succeed in fetching JAMA’s DOI to improve the worst error in the evaluation set when the LLM was off.

The credibility system still fails because the scoring system is additive and not statistical, which led to the NEJM article being scored a credibility of 1.0, capped by `_combine_signals`. There is also no confidence interval on any individual score; thus, the system does not provide the user with how sure it is about a particular source.

With the LLM layer off, the worst single error stayed the same as the baseline (0.410), and no improvements made in the credibility system were able to improve upon this. Band accuracy did improve slightly, but I believe there's more room for the rules-only layer to be more reliable on its own before falling back on an LLM to improve context. Additionally, this credibility system does not support the *haiku-4-5* model, as it silently breaks the LLM layer. The official fix was not pulled into this project's version.